

Aryan Singh

☎ +91 7985447278 | ✉ aryansingh81104@gmail.com | 🔗 LinkedIn | 🐙 GitHub

SUMMARY

I am a results-driven student at VNIT Nagpur with a strong focus on backend systems, full-stack development, and scalable architecture. I build high-performance solutions ranging from custom C++ database engines to production-ready web platforms. With experience leading technical initiatives, I excel at translating complex engineering challenges into efficient, user-centric systems while driving automation and impact at scale.

EXPERIENCE

Head of Technical Affairs E-Cell, VNIT

May 2024 – May 2025

Executive Body Member

Nagpur, MH

- **Digital Transformation:** Spearheaded the overhaul of legacy ticketing processes by building an automated online booking system with QR-based entry scanning, **reducing event queue wait time by 60%**.
- **User Experience Optimization:** Enhanced end-to-end user engagement by **45%** through streamlined registration workflows and high-performance internal automation tools.

PROJECTS

AtlasORM Custom C++ ORM Engine | C++, SQLite, STL, System Design [GitHub](#)

- Engineered a custom Object-Relational Mapping (ORM) engine in C++ with a layered architecture, abstracting SQL complexity into clean, reusable object-oriented code.
- Developed a dynamic query builder supporting method chaining (`.where`, `.orderBy`, `.limit`) for type-safe and flexible data retrieval.
- Implemented efficient DB-to-object mapping and Repository patterns to optimize scalable database interactions.

DocChat AI PDF Q&A and Summarizer | Python, Gemini API, Sentence-Transformers [GitHub](#)

- Built an AI-powered document assistant utilizing semantic search and LLMs to provide context-aware answers and summaries from unstructured PDF data.
- Optimized text extraction pipelines and generated semantic embeddings to map user queries to relevant content via cosine similarity.

Consonite Live Animated Event Portal | React.js, GSAP, Node.js, MongoDB [GitHub](#) | [Live](#)

- Architected the official portal for a major campus event, featuring a high-performance frontend with GSAP animations and a fully responsive interface.
- Developed a robust Node.js and Express backend to manage high-volume concurrent user registrations and ensure data persistence.

VidSocial Microblogging Platform | Node.js, Express, MongoDB, Cloudinary [GitHub](#)

- Developed a hybrid social platform featuring adaptive video streaming via Cloudinary integration and a secure microblogging feed.
- Implemented secure infrastructure using JWT-based authentication and custom middleware to protect REST API routes.

TECHNICAL SKILLS

Languages: C++, C, Python, JavaScript, SQL (PostgreSQL, MySQL, SQLite), HTML/CSS

Frameworks & Libraries: React.js, Node.js, Express.js, Tailwind CSS, GSAP, Sentence-Transformers

Developer Tools: Git, GitHub, Postman, Vercel, Gemini API, Cloudinary, JWT, REST APIs

Relevant Coursework: Operating Systems, DBMS, Object-Oriented Programming, Computer Networks

EDUCATION

Visvesvaraya National Institute of Technology

Bachelor of Technology

Aug. 2023 – Present

Nagpur, Maharashtra